WISP: Printable Graphene-Based Wearables for Force-Based Micro-Gesture Recognition

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Abstract

This paper introduces WISP, a printable graphene-based wearable system that enables truly imperceptible micro-gesture recognition with sub-10mm finger movements—up to 4× smaller than existing approaches. While conventional gesture systems require conspicuous hand motions that disrupt social interactions, WISP detects subtle pinch-like micro-gestures that are virtually invisible to observers yet produce distinct force signatures at the wrist through tendon and skin deformation. Our co-designed hardware-software pipeline leverages Laser-Induced Graphene (LIG) technology to create ultra-thin, skin-conformal strain sensor arrays that can be fabricated on-demand using standard laser cutters and achieves $100\mu V pp$ noise floor while consuming only $10mW$ power. Evaluation with 19 participants demonstrates 99.2% hotword detection, 90.2% user-specific recognition, and 82.9% cross-user generalization. WISP maintains high accuracy across real-world conditions including walking and various clothing scenarios, scales to 12-gesture vocabularies, and extends to context-aware object interactions with 93.8% accuracy.

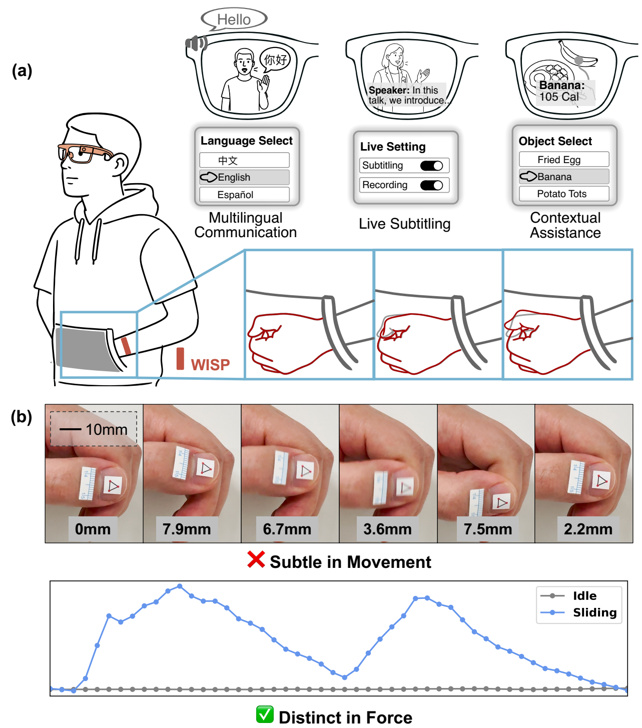

Figure 1: (a) Interaction with AI assistants in an unobtrusive manner with hands concealed. (b) Micro-gestures are subtle and hard to detect in terms of position, but have distinct and distinguishable force patterns (images were taken at a close range of approximately 10cm).

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Hardware** → **Emerging interfaces**; • **Computer systems organization** → **Sensors and actuators**; • **Applied computing** → *Physical sciences and engineering*.

Keywords

Human-Computer Interaction, User Interface, Gesture Recognition, Sensor Design and Fabrication, Laser Induced Graphene, Low Power Circuit

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1 Introduction

AI-powered smart glasses and wearables are transforming everyday social interactions, bringing AI assistants directly into conversations and daily activities. From real-time instant language translation enabling seamless multilingual communication [7, 34] to real-time subtitling [13, 17, 44, 47, 52] and contextual assistance [5, 16], these applications demand interaction modalities that are socially appropriate and completely unobtrusive. Unlike traditional computing scenarios where users can step aside to interact with devices, AI-augmented social interactions require continuous, discreet input methods that preserve natural social dynamics.

This need for discretion has driven gesture recognition research toward increasingly subtle interaction paradigms. Early systems relied on large arm and body movements clearly visible to observers (e.g. [42]), while recent advances have progressed through coarse-grained hand gestures (e.g., American Sign Language with roughly 5-10cm displacement) [33] to fine-grained finger movements (1-5cm displacement) [3, 24, 45]. However, even these "fine-grained" gestures remain conspicuous in social settings [54] – finger swiping motions or deliberate hand positioning are too obvious and disrupt natural conversation flow.

The next frontier requires gestures so subtle they can be performed with hands concealed—in pockets, under tables, or naturally at one's side—while maintaining reliable recognition accuracy. This represents a fundamental shift toward truly imperceptible micro-gestures with sub-10mm overall displacement. However, to achieve this level of imperceptibility, we cannot rely on familiar finger-swipes and tap gestures, which remain inherently obtrusive. Instead, we must explore fundamentally different gesture paradigms.

One particularly promising approach leverages subtle finger force variations during natural pinching motions between thumb and fingertip—movements that preserve the rich expressiveness of finger interaction while remaining virtually undetectable to observers. However, existing pinch-based approaches like Gaze and Pinch gestures proposed for VR [39, 40] require fingers held out in front of the VR headset and are not appropriate for natural interaction with AI assistants, which need hidden-hand scenarios where hands remain in pockets or concealed, precluding line-of-sight sensing approaches entirely.

Current approaches to fine-grained gesture recognition face significant practical limitations that prevent deployment in truly unobtrusive scenarios. Vision-based systems (e.g. [2]) require line-of-sight and fail when hands are occluded, such as when operating devices in pockets or under clothing. Acoustic approaches (e.g. [24]) similarly suffer from occlusion sensitivity and are limited by tracking resolution (9.0mm). More invasive solutions like instrumented gloves [46] provide excellent sensing capability but are impractical for daily use due to their obtrusiveness and social acceptability concerns.

Our Approach. While ring-based systems can detect these subtle gestures through direct finger instrumentation, we introduce a complementary wrist-based approach that exploits a key physiological insight: finger movements, particularly subtle pinching forces, create measurable deformations in wrist tendons and surrounding skin tissue. Rather than attempting to directly capture diminutive finger motions, our system detects these indirect manifestations through high-sensitivity strain measurement at the wrist using printable, ultra-thin sensors. This wrist-based paradigm has several unique benefits, such as leaving fingers completely unencumbered for daily activities, enabling operation in non-line-of-sight scenarios (hands in pockets, under sleeves), and allowing potential integration into existing smartwatch form factors.

Our system, WISP[1], utilizes flexible, printable strain sensors that conform to the wrist skin surface, positioning them optimally to capture tendon and skin deformations linked to finger force patterns. Printability is essential for making this approach practical—our skin-conformal sensors function like medical patches that naturally degrade with repeated flexing, requiring periodic replacement. The emerging Laser-Induced Graphene (LIG) approach that we employ enables ultra-low-cost, environmentally friendly production that makes this model economically viable while eliminating the chemical waste of conventional sensor fabrication.

Figure 1 illustrates how this gesture works, showing how visually nearly indistinguishable pinch movements that involve displacement well under 10mm can still produce distinctive force signatures at the wrist that can be leveraged for interaction.

Technical Challenge. Realizing sub-cm gesture detection from the wrist presents several critical design constraints that traditional sensing approaches cannot simultaneously satisfy. These requirements create a challenging three-way tradeoff between form factor, cost, and performance.

First, the system must maintain an extremely low profile to achieve skin-conformal flexibility necessary for comfort and accurate sensing. Second, the conformal skin contact required for effective tendon measurement causes sensors to experience mechanical wear with finite reuse cycles, necessitating low-cost fabrication for practical replacement. Third, detecting subtle tendon deformations corresponding to sub-10mm finger movements requires exceptional sensitivity typically achieved only through expensive commercial strain sensors—yet our application demands flexible, low-cost, disposable designs that traditionally lack sufficient sensitivity for such minute skin deformations.

Rather than pursuing sensitivity improvements solely through material innovations, we resolve this challenge through integrated hardware-software co-optimization combined with targeted modifications to LIG sensor fabrication. We achieve skin-conforming flexibility by reducing substrate thickness from a typical 75μm used in prior LIG fabrication work to 25μm, while maintaining sensor conductivity and durability through careful fabrication optimization. Rather than relying on a single high-performance sensor, we employ a multi-sensor array design that provides spatial diversity, redundancy against fabrication variations, and richer signal patterns for gesture classification. Our co-designed hardware-software pipeline achieves a $100\mu V_{pp}$ noise floor at the analog stage while consuming only 10mW, enabling practical all-day operation in an ultra-low-profile, skin-conformal form factor.

Performance. In head-to-head comparisons, WISP outperforms Electromyography (EMG) sensors and accelerometers in Signal-to-Noise-Ratio (SNR) in both static and dynamic conditions for micro gesture detection. Our evaluation with 19 participants demonstrates robust recognition rates: 99.2% accuracy for hotword detection, 90.2% for user-specific models, and 82.9% for cross-user generalization. The system scales to a 12-gesture vocabulary through sequential combinations while maintaining 85.1% accuracy.

Real-world testing across varied conditions reveals strong robustness—accuracy remains high when wearing gloves, during different arm orientations, and in walking—scenarios that typically

[1]WISP = Wearable Imperceptible Sensing Platform

challenge gesture recognition systems. We further extend functionality through object detection, achieving 93.8% accuracy in identifying four common objects, with gesture recognition remaining stable across different interaction surfaces. This context-aware capability enables richer, object-specific gesture interpretations without additional hardware.

To summarize, our main contributions include:

- We introduce imperceptible micro-gesture sensing with sub-10mm (2.8mm – 9mm) overall displacement that exploits skin surface deformation patterns, thereby enabling gesture detection at scales previously thought impossible without wearing anything on the fingers.
- We design a fabrication process that achieves the challenging balance of sensor flexibility, sensitivity, and cost-effectiveness through ultra-thin LIG sensors and multi-channel arrays.
- We develop a hardware-software pipeline that achieves 100μV (peak-to-peak) noise floor at 10mW power consumption, making high-fidelity wristband sensing practical for continuous wear.
- We demonstrate robust performance across diverse real-world conditions, and further achieve 90.2% accuracy for personalized recognition with an additional extension to context-aware object interactions in multi-user studies.
- We provide an open-source implementation with Arduino-compatibility to accelerate future research in this area.[2]

2 Background

2.1 Gesture Recognition in XR

The evolution of Extended Reality (XR) technologies has fundamentally transformed interaction paradigms. AR/VR headsets like Meta Quest 3, Microsoft HoloLens 2, Magic Leap 2, and Apple Vision Pro employ optical hand tracking and in some cases, gaze+pinch interaction [2, 39, 40], where users glance at UI objects with their eyes and pinch with their fingers to activate them. However, these approaches share a fundamental limitation: they require line-of-sight visibility between optical vision sensors and hands.

The transition from controlled AR/VR environments to everyday smart glasses exposes a critical gap: *social acceptability.* Gesture vocabularies that are effective for AR/VR headsets, such as gaze+pinch, air-tapping, or extended hand positioning, disrupt natural social dynamics essential for AI-augmented conversations [22, 34], real-time translation [7], or contextual information systems [5].

Thus, the next generation of XR applications demands gesture recognition that operates under fundamentally different constraints: (1) *occclusion tolerance* – recognition must function when hands are in pockets, under tables, or naturally at one's side; (2) *social discretion* – gestures should be virtually undetectable to observers; (3) *continuous operation* – systems must function during walking, conversation, and daily activities.

[2]https://github.com/leizhenyu97/WISP

2.2 Fine-grained hand-gesture recognition

Hand gesture recognition has evolved over the years from *coarse-grained* gestures (> 50mm, e.g. ASL) motion recognition to *fine-grained* gestures (10 – 50mm, e.g. fingertip swiping across joints).

Table 1 summarizes recent developments in hand gesture recognition systems. Armband and wristband-based methods [9, 20, 21, 24, 30, 33, 42, 48, 56, 57] are typically limited to sensing coarse-grained whole-hand gestures due to their distance from the finger, including systems employing indirect sensing from the wrist, such as WristFlex [9] (pressure-based) and Tomo [56] (impedance-based). They distinguish only gestures with movements substantially exceeding 10mm due to sensing modality and sensitivity constraints. To further improve the granularity, several ring-based methods have been proposed [3, 6, 6, 45, 49, 54] but they remain above 10mm-level and interfere with natural finger use.

Several systems approach but do not achieve sub-10mm imperceptible gesture recognition. EchoWrist [24], achieves 9mm fingertip tracking sensitivity through ultrasound from a wrist-based device; however, it relies on line-of-sight acoustic sensing. Thumb-In-Motion [3] and Ring-a-Pose [54], have explored "micro gestures" through swipe-based interactions; however, these involve movements with tracking resolution of 19.9mm resolution (fingertips) that remain visible to observers. Pyro [15] introduces another micro gesture vocabulary but requires optical line-of-sight and depends on detecting visible position changes. Soli [26] demonstrates impressive mm-wave radar capabilities for miniature gestures but requires hands positioned directly in front of the sensor. Meta's EMG wristband [35] shows promise for wrist-based sensing, though public demonstrations have focused primarily on coarser movements such as finger swipes, tap-and-hold with different fingers, and handwriting based on wrist motion.

In contrast, WISP achieves truly imperceptible sub-10mm gesture recognition by detecting force-based patterns at the wrist that differentiate gestures performed in nearly identical spatial positions (e.g., double-press versus press-and-hold) without requiring line-of-sight or visible hand positioning. Beyond resolution, existing approaches also fall short on other metrics such as power efficiency, user comfort, and cost as shown in Table 1.

2.3 Graphene-based strain sensing

From a sensing perspective, our work is based on graphene-based strain sensors, which is different from most existing wearable gesture recognition solutions. As shown in Table 1, existing approaches predominantly use capacitive [3, 42, 48], impedance [49, 56, 57], EMG [30, 33], acoustic [20, 24, 54], proximity [45], electric field [6], and piezo [9, 21] sensing.

In our work, we leverage graphene's unique properties, including its electrical conductivity [4], flexibility [12], mechanical strength [55], and biocompatibility [41], which make it an excellent candidate for high-performance strain sensing. Importantly, laser-induced graphene (LIG) technology transforms graphene into a practical solution—enabling direct fabrication of custom sensor arrays on flexible substrates using only a standard laser cutter [27, 51], making this approach scalable, customizable, and low-cost.

	Gesture Set	Form Factor	Sensor Customizability	Low-power ($\sim 10mW$)?	Sensing Method
Capband [48]	Coarse-grained	Thin wristband	✗ FPCB Fabrication	✓ $\sim 10.38mW$	Capacitive
Kim et al [21]	Coarse-grained	Sensor only	✓ Laser Engraving	✗ Not mentioned	Piezo (Strain)
Tomo [56, 57]	Coarse-grained	Thin wristband	✗ Copper foil cropping	✗ $50mW$ except BLE	Bio-impedance
GestureWrist [42]	Coarse-grained	Thin wristband	✗ Not mentioned	✗ Not mentioned	Capacitive
Neuropose [30]	Coarse-grained	Bulky armband	✗ COTS	✗ Not mentioned	sEMG
WristFlex [9]	Coarse-grained	Thin wristband	✗ COTS	✗ $96.0mW$	Piezo (Pressure)
EMpress [33]	Coarse-grained	Bulky wristband	✗ COTS	✗ Not mentioned	Piezo + sEMG
Interferi [20]	Coarse-grained	Bulky armband	✗ COTS	✗ $\sim 2W$	Bio-acoustic
EchoWrist [24]	Coarse-grained + Tracking (MFTEDE: $9.0mm$)	Thin wristband	✗ COTS	✗ $57.9mW$	Acoustic
Thumbtrak [45]	Fine-grained	Ring	✗ COTS	✗ $120mW$	Proximity
Thumb-In-Motion [3]	Fine-grained	Ring	✗ FPCB Fabrication	✗ $475mW$	Capacitive
EFRing [6]	Fine-grained + Tracking (MFTEDE: N/A)	Ring	✗ Copper foil cropping	✗ Not mentioned	Electric Field
Z-Ring [49]	Fine-grained	Ring	✗ FPCB Fabrication	✗ $2.4W$	Bio-impedance
Ring-a-Pose [54]	Fine-grained + Tracking (MFTEDE: $19.9mm$)	Ring	✗ COTS	✗ $148mW$	Acoustic
WISP	**Micro (mm-level max; sub-mm-level sensitivity)**	**Thin wristband**	**✓ Laser Engraving**	**✓ $10.13mW$**	**Graphene-based Strain**

Table 1: Comparison of related work vs WISP. COTS: Commercial Off The Shelf; MFTEDE: Mean FingerTip Euclidean Distance Error (calculated from individual fingertip results if not provided directly).

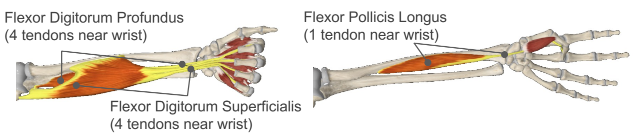


Figure 2: Anatomical example of muscle and tendon dynamics during finger actions.[3]

However, prior work on graphene-based strain sensors has primarily focused on individual strain sensors with controlled laboratory demonstrations, rather than multi-user deployment studies with real-world evaluation. Luo et al. [32] fabricated LIG strain sensor arrays but did not address the challenges of 2D routing and wearable integration. Yang et al. [53] developed a multi-modal LIG sensor combining strain, temperature, and chemical detection capabilities, but used only a single strain sensing element without system-level optimization for specific applications.

3 WISP System Overview

This section provides a high-level overview of WISP's design philosophy, key insights, and system architecture before diving into implementation details.

Design Philosophy and Key Insights. WISP is built around three fundamental insights that enable imperceptible micro-gesture recognition:

Physiological Insight: Indirect Force Measurement. Rather than attempting to directly capture diminutive finger motions at the fingertips, WISP exploits a key physiological principle: finger movements, particularly subtle pinching forces, create measurable deformations in wrist tendons and surrounding skin tissue. Figure 2 illustrates the anatomy of a male left forearm with the palm facing upward. Nine flexor long tendons, located on the palmar side and traversing the wrist, are responsible for finger flexion: Flexor Pollicis Longus for the flexion of the thumb, while Flexor Digitorum Superficialis and Flexor Digitorum Profundus tendons for the other four fingers. In addition, three other tendons (Flexor Carpi Radialis, Flexor Carpi Ulnaris, and Palmaris Longus) control wrist movement and also contribute to the wrist deformation captured by our sensor, since pinch-based gestures involve coupled finger and wrist behaviors. This paradigm enables unobtrusive sensing of finger movements at the wrist while maintaining high sensitivity and allowing natural, unrestricted finger movement.

Material Insight: Printable High-Performance Sensing. Traditional high-performance strain sensors capable of detecting small deformations are expensive, rigid, and unsuitable for skin-conformal applications. WISP leverages Laser-Induced Graphene (LIG) technology to create flexible, ultra-thin sensors that can be fabricated on-demand using only a standard laser cutter and inexpensive polyimide substrates. This approach enables the manufacturing of disposable, skin-conformal sensors with good sensitivity.

System-Level Insight: Multi-Channel Signal Diversity. Single-point sensing at the wrist provides limited signal diversity for reliable classification of subtle gesture variations. WISP employs a multi-sensor array design that provides spatial diversity, redundancy against fabrication variations, and richer signal patterns essential for distinguishing micro-gestures.

Technical Challenges and Solutions. Achieving reliable sub-10mm gesture detection from the wrist presents three fundamental challenges that WISP addresses through integrated hardware-software co-design:

Challenge 1: Skin-Conformal Flexibility vs. Sensitivity Trade-off. Traditional strain sensors are either highly sensitive but rigid, or flexible but lacking sensitivity. WISP's *Printable Sensor Array Subsystem* (§4) employs ultra-thin skin-conformal substrate design ($25\mu m$ vs. typical $75\mu m$+ in LIG fabrication) combined with a 2×2 array of LIG-based strain sensors. Multi-directional sensitivity and durability under repeated use are further ensured by various design- and fabrication-level techniques.

Challenge 2: Low-Cost Fabrication vs. High Performance. Commercial high-performance strain sensors are expensive and unsuitable for disposable applications. WISP's *High-Sensitivity Readout Subsystem* (§5) enables custom sensor arrays at <$0.20 per unit through LIG fabrication, while achieving $100\mu V$ peak-to-peak noise floor via quarter Wheatstone bridge configurations, 4th-order analog filtering, and 24-bit Sigma-Delta ADC.

Challenge 3: Noise vs. Power Consumption Trade-off. Detecting subtle signals typically requires high-power analog conditioning. WISP's *High-Sensitivity Readout Subsystem* (§5) achieves 10.13mW total power through optimized analog filtering and duty-cycled operation. WISP also employs an efficient processing pipeline for further signal processing, feature extraction, and gesture classification. Our end-to-end implementation is described in §6.

[3]Image courtesy of Complete Anatomy [1] with customized annotation.

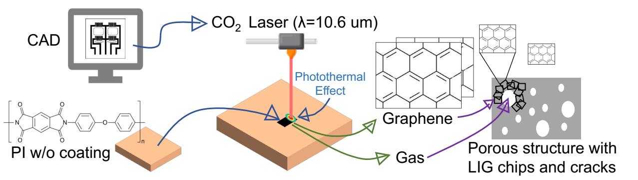


Figure 3: Schematic illustration of the LIG fabrication process, showing CO_2 laser patterning of polyimide (PI) substrate to form conductive graphene structures.

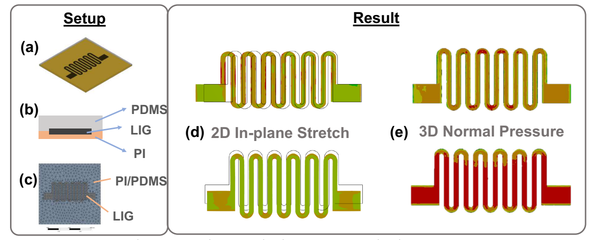


Figure 4: Simulation-based design validation: *FEM setup:* (a) CAD pattern; (b) Cross-sectional view of the sensor's internal structure; (c) Meshing. *Result:* (d) 2D stretching and (e) 3D bending induce distinctly different strain responses in orthogonal directions (i.e. along x and y axes).

The following sections detail the design, implementation, and evaluation of each of these subsystems.

4 Sensor design

LIG has emerged over the past decade as a promising sensor fabrication technology. Through a single, computer-controlled Direct Laser Writing (DLW) step on the Polyimide (PI) substrate, piezoresistive sensors with performance comparable to commercial strain gauges can be easily synthesized, much like printing with an inkjet. This fabrication process is inherently customizable and economical for disposable applications. However, despite LIG's growing popularity within the materials and sensor communities, its translation to application-oriented, end-user systems, as mentioned in Section 3, remains largely unsolved.

In this section, we first discuss why reducing the thickness enhances *skin-conformability* (§4.1, §4.2) and how this reduction consequently introduces challenges in *sensitivity* and *durability*. Then, §4.3 describes our strategies to address these challenges through single sensor design and fabrication, interference decoupling, and robustness enhancement.

4.1 Fundamentals of Laser Induced Graphene

LIG Synthesis Principle. As Figure 3 depicts, the LIG fabrication involves only a CO_2 laser cutter and a carbon precursor substrate, Polyimide (PI). The process is solely based on the photothermal effect during laser engraving: PI absorbs energy from the infrared CO_2 laser beam, generating internal lattice vibrations. This induces a highly localized high temperature that breaks the molecular bonds within PI, accompanied by gas release. This released gas suppresses the PI decomposition before graphitization, promoting larger carbon cluster formation with minimal oxidation [10, 27]. The result is a piezoresistive graphene layer with a unique porous structure and cracks, where the contact resistance changes dramatically under external force.

Advantages of LIG. LIG enables precise, single-step conversion of carbon within the substrate to conductive graphene, bypassing complex and hazardous chemical steps. It offers scalability and customizability, as sensor designs can be modified by simply tuning CAD patterns. Compared to screen or inkjet printing, LIG is mask-free, uses a single laser cutter for both engraving and cutting, and is low-cost due to affordable laser equipment (<\$10k) and inexpensive substrates (<\$0.2 each) without requiring nanoparticle inks. The resulting sensors in this work are small ($3cm \times 3cm \times 300\mu m$ with PDMS coating) and lightweight (0.14g without wires), making them attractive for wearables.

4.2 Flexible and Conformal Sensor Design

While LIG is a promising technology for our use case, LIG synthesis on polyimide (PI) films in previous works typically involved thicknesses over 75 μm [27, 53], leading to rigidity unsuitable for skin conformance. According to the Euler–Bernoulli Beam Theory [25], flexural rigidity (or bending stiffness) is determined by:

$$\text{Flexural Rigidity} = EI = \frac{1}{12}Ebh^3 \tag{1}$$

where E represents Young's modulus, I the second moment of area, b the width, and h the thickness. Given that flexural rigidity scales with the cube of thickness, even small reductions yield significant improvements in flexibility. Therefore, we reduced the PI thickness from 75 to 25 μm. With a PI modulus (2.76 GPa) much higher than that of human skin's (1 MPa) [29, 50], this thickness reduction achieves a conformal fit to the skin.

In the approach described above, we prioritized low cost and high comfort in the tradeoff between cost, comfort, and performance by utilizing LIG technology with ultra-thin, skin-conformal substrates. This, however, makes achieving high performance challenging. First, thinner PI films tolerate lower laser fluence during fabrication to prevent burn-through, leading to incompleteness in the graphenization of the sensor. Second, these films also exhibit less mechanical strength and are more susceptible to external interference, including scratching and tearing. These challenges are addressed through sensor- (remainder of this section) and system-level optimizations (Section 5).

4.3 From Single Sensor to Robust Array

1. CAD Design Guided by Simulation. While traditional strain sensors are typically unidirectionally sensitive, we require sensors that provide distinct and differing responses to strain in various directions, enabling multi-directional sensing within a single device. Furthermore, a miniature size is necessary due to the restricted wrist area. With Finite Element Method (FEM) simulation, we iteratively explored various geometric designs and ultimately selected an optimized zigzag-with-curve pattern, inspired by example designs in [23]. This features a 0.25mm linewidth and a 0.5mm arc radius, taking the spatial resolution of 0.254mm in the laser cutter into consideration.

Figure 4 (a-c) depicts the CAD, internal structure, and element meshing used in the simulation setup. Unlike previous works that involved only LIG, our simulation method supported the entire sandwich structure of the sensor, ensuring the mutual interactions of materials with different properties were accurately considered. Figure 4(d,e) validates our design, showing strain distribution within sensors under both 2D stretching (in-plane) and 3D bending (out-of-plane), each with two orthogonal settings. This indicates that the zigzag-with-curve design can provide distinct anisotropies for different directions and therefore is suitable for our application.

2. Laser Fabrication Parameters. In laser-based graphene synthesis, controlling parameters, such as power, speed, and Pulses Per Inch (PPI), directly impact the laser's energy density (fluence), influencing both conductivity and structural integrity. We optimized these parameters to ensure graphene formation on the thinner PI substrate without damage.

As depicted in Figure 5(b), the sensor remains non-conductive at lower power settings (e.g., 4%) and higher speeds (e.g., 6%). Increasing the power or reducing the speed darkens the pattern, leading to decreased resistance and enhanced conductivity. It is important to note, however, that continuing this trend will ultimately lead to sensor damage (e.g. 6% power and 4% speed). Optimal settings (5% power and 5% speed) prevent substrate burn-through while ensuring stable conductivity.

3. Addressing Coupling Issues. In any multi-sensor array, coupling between components can introduce interference and degrade signal quality. We addressed three forms of coupling:

- *Sensor-to-sensor coupling:* To minimize strain transfer between adjacent sensor units, we introduced strategic cuts between them, as shown in Figure 5(c). These cuts create mechanical isolation, reducing deformation propagation from one sensor to its neighbors.
- *Sensor-interconnect coupling:* The close proximity between sensors and their interconnects can lead to interference, where resistance changes in the interconnect paths contaminate sensor readings. We addressed this through interconnect geometry optimization by widening the traces to reduce their contribution to overall resistance and thereby mitigate interference, while carefully balancing the layout so that these wider interconnects do not reduce array density due to space limit.
- *Interconnect-wire coupling:* Motion at the LIG-copper wire junction can introduce significant noise artifacts. We addressed this through a multi-layer stabilization approach with silver paste and epoxy (details in §6).

4. Enhancing Sensor Robustness. To ensure our sensor array can withstand real-world usage conditions, we implemented several robustness enhancements. We first spin-coated a layer of PDMS solution on LIG to prevent graphene detachment due to external friction or scratching, while maintaining the sensor's flexibility due to the elastomeric properties of PDMS. We then applied round-cornered cutting, instead of normal rectangular cuts, to distribute stress more evenly during deformation. This enhanced the structural integrity and lifespan of the sensor array. Finally, a medical tape layer was incorporated between the PI substrate and the skin. This tape [19] withstands >1kg shear force for 300+ hours with 3.1 psi loop tack, demonstrating strong adhesiveness and reliable strain transfer throughout our experiments, while retaining biocompatibility and breathability.

5 Readout Design

The readout system's design aimed for high sensitivity to detect minute wrist-area deformations and low power consumption for extended battery life. This necessitated low-level optimizations in signal enhancement, interference suppression, noise reduction, and power management. Through several iterations (final version shown in Figure 6 and 7(b)), our system achieves a noise floor below $100\mu Vpp$ with a total power consumption of only $10.13mW$.

Signal Enhancement. To improve the signal-to-interference-and-noise ratio (SINR), many systems amplify signals after filtering out the DC component to prevent saturation. However, our application requires preserving valuable near-DC signal information while eliminating unnecessary fixed offsets. We accomplish this using a quarter Wheatstone bridge configuration with pseudo-differential inputs. The positive arm captures both signal and offset, while the negative arm contains only the offset. Subtracting these components theoretically removes the offset while preserving near-DC signals of interest. We then further amplify the resulting signal using a Programmable Gain Amplifier (PGA). In practice, sensor fabrication variations and resistor tolerances introduce residual offset, which limits our maximum amplification gain.

For signal digitization, common Successive Approximation Register (SAR) ADCs (usually 8-12 bits) lack sufficient sensitivity for micro gesture detection. Instead, we implement a single 24-bit Sigma-Delta (Σ-Δ) ADC architecture, providing superior precision. To scale this to multiple channels, an analog multiplexer with low on-resistance is synchronized via the microcontroller. To prevent inter-channel crosstalk after channel switching, the ADC's digital filter register is reset under firmware control before new conversions.

Mitigating Interference. The next challenge we addressed is suppressing interference originating from external sources, such as powerline interference and its harmonics. WISP experiences channel imbalances due to sensor inconsistencies introduced by the laser-printing process, causing common-mode noise to convert into differential noise. To mitigate this interference, we employed a 4th-order Chebyshev Type 1 Low-Pass Filter (LPF) for its sharp transition between the passband and stopband with minimal stages. This configuration achieves -62.5 dB attenuation at 60 Hz with only two stages. The Sallen-Key filter architecture enhances the filter's robustness against component tolerance variations, ensuring that real-world performance closely aligns with simulation.

We used analog filtering since high-frequency harmonics could alias into near-DC frequencies after A/D conversion, making software filtering less effective. Furthermore, intermodulation and powerline bandwidth extension, resulting from sampling jitter and nonlinearity, are coupled into the system as well. Digital filters, such as notch filters, cannot provide sufficient filtering without analog conditioning. To further suppress interference from leakage, the LPF itself, the Multiplexer (MUX), and the Bluetooth Low Energy (BLE) module, we introduced an Anti-Aliasing Filter (AAF) before the ADC to filter out unwanted frequencies.

Suppressing Noise. Next, we looked to minimize intrinsic system noise. The Σ-Δ ADC architecture can efficiently mitigate quantization noise through oversampling and decimation, which shifts this noise to higher frequencies. Low-pass digital filters

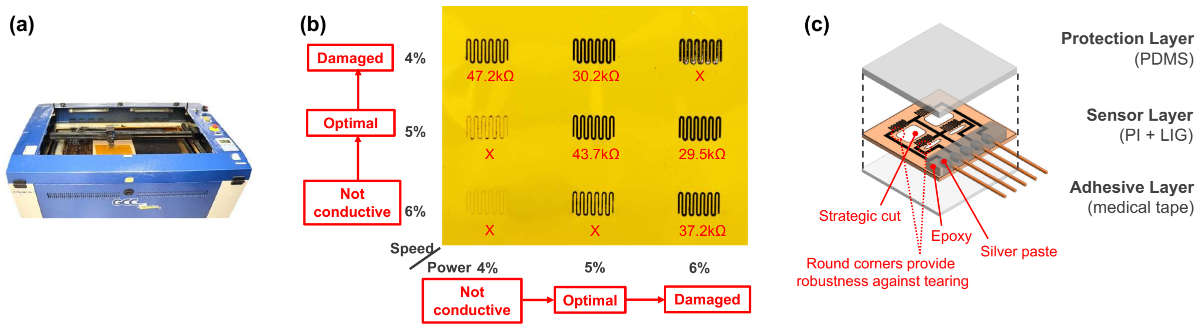


Figure 5: Sensor Fabrication: (a) Laser system used for LIG synthesis. (b) LIG sensor patterns produced under varying laser settings. (c) Exploded view of the WISP sensor array structure.

(Sinc4+Sinc1) after A/D conversion further remove these high-frequency components, eliminating most quantization noise. Furthermore, the combination of analog LPF, AAF, and digital filters also restricts the system bandwidth, thus achieving broadband noise reduction.

Low Power Design. Achieving high accuracy, low noise, and low power simultaneously presents significant challenges. To achieve this, we selected a single low-power Σ-Δ ADC, combining with a standalone external analog switch to enable channel multiplexing. This replaces the need for simultaneous sampling with multiple ADCs while maintaining precision. For the multi-stage LPF, low-power, zero-drift operational amplifiers were chosen to prioritize precision and power. While, as a tradeoff, it generates extra high-frequency noise due to internal auto-zero modulation, we further remove it through the subsequent AAF.

In the digital domain, MCU and BLE transmission power is non-negligible. Default BLE settings (27-byte MTU, low-bandwidth PHY) are inefficient. To optimize, we aggressively reduced the BLE transmission time by renegotiating the transmission protocol to boost the MTU to 247 bytes with a 2MHz PHY layer. Therefore, the MCU can be duty-cycled more effectively, with longer sleep periods, to save power.

Open-source with Arduino-compatibility. We open-source the design for the sensors, circuits, and firmware. Regarding the components used in our circuit, although most of their drivers were originally developed for other platforms, we customized them to ensure seamless integration and flexibility within the Arduino environment.

6 Putting Things Together

Sensor Design and Fabrication. For the simulation, we developed an FEM simulation environment with Ansys Mechanical, configuring material properties based on established values [18, 36, 37]: PDMS (Young's modulus: 1.32MPa, Poisson ratio: 0.495), PI (2.76GPa, 0.34), and LIG (200MPa, 0.1). We set the thickness of PDMS, PI, and LIG to be $250\mu m$, $25\mu m$, and $50\mu m$, respectively, with $15\mu m$ of LIG embedded in PI substrate to match real-world printed LIG characteristics, as the simulation input.

The fabrication process consists of three main phases:

(1) *Laser-cutting and engraving:* We use a LaserPro Spirit GLS (CO_2, 80W, $\lambda = 10.6\mu m \pm 0.03\mu m$) to process PI substrate (Kaneka 100NP; Apical). To prevent substrate deformation during processing, we secure the PI to 1/8-inch acrylic boards. Initial cutting is done in vector mode (20% power, 10% speed, and 400 PPI), followed by LIG engraving in raster mode (5% power, 5% speed, and 400 PPI), and finally strategic cutting with the same cutting parameters.

(2) *Protective coating:* A PDMS layer (Sylgard 184, mixed at 10:1 ratio) is spin-coated on the sensor array at 400 rpm for one minute.

(3) *Interconnect stabilization:* Electrical connections are established using silver paste and baked at $80°C$ for one hour. Then, epoxy is applied to reinforce the connection. Finally, ultrathin, flexible medical tapes (TM8500; Mactac) are placed under the sensor for skin adhesion.

Signal Acquisition Pipeline. For signal conditioning, a voltage divider is first used to turn the resistance of the sensor to a voltage signal. Then, a quarter Wheatstone bridge is used to remove the signal baseline (reference resistors: $51k\Omega$, 0.1% precision). For each of the 4 channels, we apply a 4th-order (2-stage) Chebyshev Type 1 analog LPF based on Sallen-Key architecture, with a corner frequency of around 14Hz. ADA4051 ICs are used as the zero-drift Operational Amplifiers for both buffering and filtering in the LPF. A multiplexer based on MAX4734 is used to switch among the 4 channels, followed by a well-matched 1st-order AAF (with ADA4051 and X2Y capacitors (Johanson Dielectrics)). The final stage uses a 4x PGA before the AD7124 $\Sigma - \Delta$ ADC with a clock speed of 614.4 kHz (full power mode) and decimated to 200 SPS (50 SPS each channel) with a Sinc4-Sinc1 post digital filter. The digitized data is read by the MCU (Nordic NRF52840 Soc integrated into a Seeed Xiao module) through SPI, and finally sent to the host via BLE. Figure 7 shows the connection and layout of different functional blocks in the hardware.

Algorithm Implementation. Our processing pipeline begins with wavelet denoising using Symlet2 mother wavelet (Level 2 approximation) and Bayes-median thresholding. After segmentation, we apply Asymmetric Least Square approximation ($\lambda = 10^5$; $p = 10^{-3}$) [11] for baseline removal. Feature extraction combines Catch22 [31] with KPCA (20 components, RBF kernel), followed by z-score normalization. Final classification uses a Random Forest (1000 estimators, gini criterion) implemented in Scikit-Learn.

Gesture Vocabulary. Our gesture vocabulary consists of four micro gestures based on a pinch interaction paradigm. For all gestures, users begin and end in the "idle" position, with the thumb tip resting on the middle of the index finger. When performing non-idle gestures, the participant applies directional force by leaning their thumb toward the required direction. As Figure 8(a) illustrates, a critical design constraint is that the thumb remains restricted to

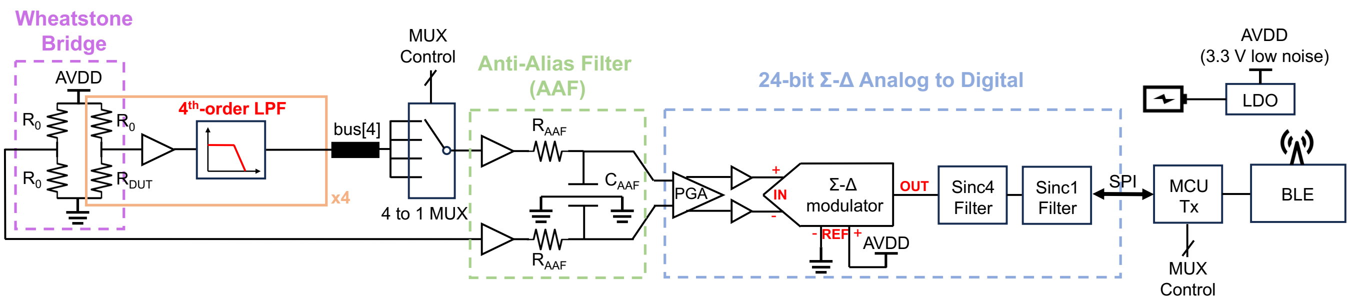


Figure 6: Comprehensive hardware architecture of WISP, enabling high sensitivity, low noise, and low power.

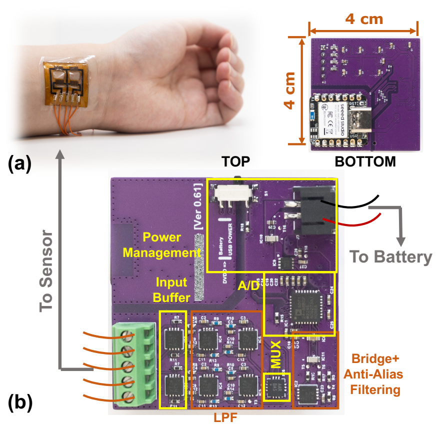


Figure 7: Hardware implementation: (a) WISP worn on the wrist; (b) Compact PCB prototype with annotation for connection and different functional blocks.

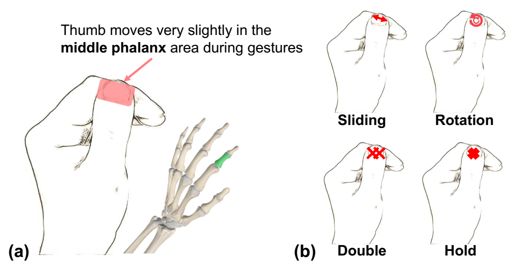


Figure 8: Our proposed force-based micro gestures: (a) Range of thumb movement during gestures, showing minimal displacement. (b) Illustration of the 4 base micro-gestures.

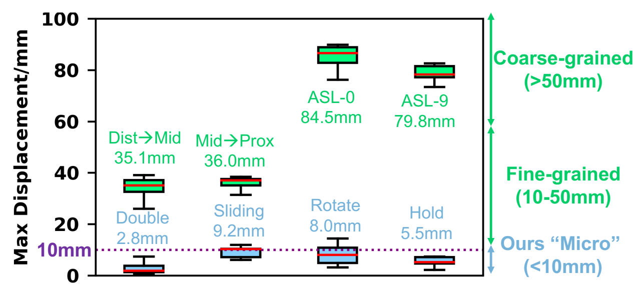


Figure 9: Gesture granularity benchmarks, comparing the *maximum overall* displacement for different gestures (Ours i.e. WISP, Fine-grained: finger swiping, Coarse-grained: ASL). We see that WISP is able to sense truly micro-gestures that are smaller than 10mm.

the middle phalanx area throughout execution, ensuring extremely minimal overall movement. Importantly, the thumb and index finger maintain continuous contact during the entire gesture sequence—eliminating visually obvious actions such as lifting, detaching, or tapping that characterize previous approaches.

Figure 8(b) details the four gestures: a) *Sliding:* The thumb leans laterally while maintaining contact; b) *Rotation:* The thumb executes a circular motion resembling a spinning top; c) *Double:* The thumb performs two rapid successive pinches without breaking contact; and d) *Hold:* The thumb applies sustained downward pressure for several seconds.

7 Evaluation

We now present experiments to evaluate WISP. In Section 7.1 and 7.2, we conducted initial single-participant benchmarking experiments on movement granularity and signal quality comparison across different modalities, as within-subject repeated measurements offer a controlled and straightforward comparison. In Section 7.3 and 7.4, we conducted a multi-user study, with Institutional Review Board (IRB) approval, to understand the usability for different tasks. We had 1710 gesture instances in micro gesture detection, 1170 instances in object recognition, and 1170 instances in the wrist joystick task, all collected in a fully randomized way. For the combined gesture task, the micro-gesture dataset was divided into training and test subsets, and pairwise class combinations were formed within each subset to avoid data contamination, resulting in 51,300 training instances. Depending on the task, each participant's session could last up to approximately one hour, yielding a comprehensive dataset across different gesture types.

7.1 Quantifying Movement Granularity

To quantify how our approach differs from existing traditional gestures, we conducted comparative measurements with two typical categories from previous work: finger swiping and American Sign Language (ASL). For finger swiping, we tracked the thumb sliding along the index finger, subdivided into two gestures (distal-to-middle and middle-to-proximal joint). For ASL, we measured transitions from ASL-5 position (i.e. relax) to ASL-0 and ASL-9 positions, recording static thumb positions.

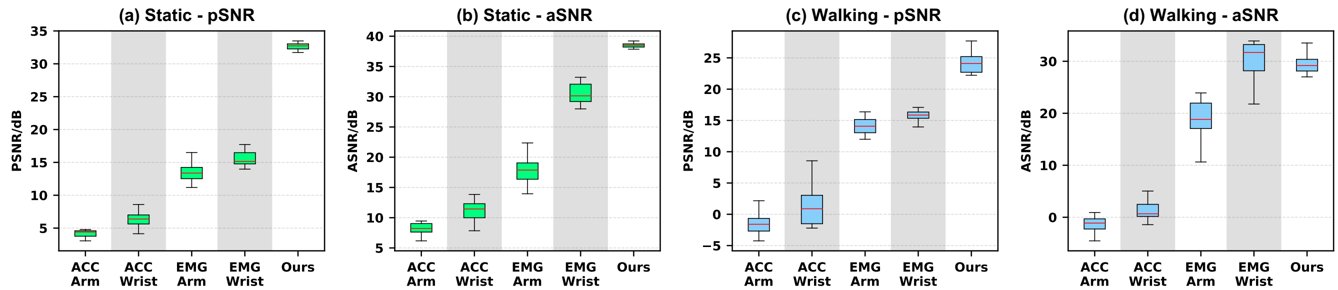


Figure 10: Comprehensive SNR comparison between our system, EMG, and accelerometer (ACC) measured on two common placement locations (wrist and forearm), two typical scenarios (static and walking), and two different metrics (aSNR and pSNR). This demonstrates WISP's superior performance across all testing conditions.

We used high-precision optical tracking to precisely measure micro gestures. We attached a sticker with three fiducial point markers to the thumbnail featuring three $1mm$-diameter dots positioned $5mm$ apart, along with a $10mm$ reference ruler. Using close-range video capture (approximately $10cm$ away), zooming in to fill the frame with the thumb, we calculated precise movement distances by measuring pixel shifts on the screen of a 14-inch Apple Macbook M1 Pro ($3024px \times 1964px$) calibrated against the reference ruler, achieving sub-millimeter resolution. We recorded measurements only at the point of maximum displacement for each gesture and calculated the average displacement of the three anchor dots.

Figure 9 presents the maximum overall displacement across all gesture types, with 320 measurements in total. The results reveal a dramatic difference—our micro gestures all remain below $10mm$ in maximum displacement, whereas conventional "fine-grained" and "coarse-grained" gestures span $30mm$ to $100mm$. This represents a 3-10× reduction in movement scale, validating our claim that WISP achieves truly imperceptible gesture recognition at previously unachievable scales.

7.2 Signal Quality Comparison

We now investigate how effectively our strain-based approach captures subtle force patterns when compared to alternative approaches, such as electromyography (EMG) and accelerometers (ACC).

Experiment Design We conducted a rigorous comparison using an OyMotion EMG armband [38] with an integrated accelerometer, testing both forearm and wrist placements. The EMG armband, with a perimeter optimized for arms, is too large for direct wrist placement. Therefore, we used firm sponge spacers to fill the gaps between the device and the wrist, ensuring consistent skin-electrode contact.

During experiments, we tested both static and motional (walking) settings to mimic two typical real-world scenarios. We also evaluated performance using two complementary metrics: power-based Signal-to-Noise Ratio (pSNR) and amplitude-based Signal-to-Noise Ratio (aSNR). The metrics were calculated as: $pSNR = 10\log_{10}\left(\frac{\sum x(t)^2}{\sum n(t)^2}\right)$ and $aSNR = 20\log_{10}\left(\frac{max(x(t))-min(x(t))}{max(n(t))-min(n(t))}\right)$. For EMG, we used the channel with the best SNR; for accelerometers, we used the RMS value across all three axes (x, y, z) as the overall acceleration measure; for WISP, we used one channel with the baseline removed, ensuring a fair comparison to EMG and ACC, which lack DC components. Overall, the experiment collected 15 instances of each micro gesture across 5 different settings in randomized order, yielding 1200 total trials.

Results. Figure 10 reveals WISP's dramatic performance advantage across all testing conditions. We summarize the results into three key insights:

(1) *Placement.* Wrist placement consistently outperformed forearm placement for all modalities, validating our wrist-worn design. This is attributed to the proximity of wrist tendons to the skin, whereas arm muscles related to fine finger movements are deeper and largely buried beneath other muscles, making them prone to interference.
(2) *Static Setting.* As shown in Figure 10(a,b), WISP provides a higher SNR than accelerometers and electromyography due to its ultra-thin and skin-conformal design. Accelerometers are typically mounted on rigid Printed Circuit Boards (PCBs) and are thus isolated from direct skin contact, leading to greater signal attenuation. Regarding EMG, its sensitivity is limited to on-wrist muscles and tendons; however, subtle finger movements also engage muscles in the hand (e.g., the Opponens Pollicis). In contrast, WISP effectively measures deformation in close proximity to skin, acquiring from both wrist-local tendons and skin stretch induced by hand movements.
(3) *Under Motion.* As shown in Figure 10(c,d), WISP maintains a high SNR under mild motion without compromising flexibility and comfort, unlike ACC or EMG. ACC-based sensing was severely affected by motion artifacts within the signal's frequency band. While EMG sensors exhibit motion robustness due to high-pass filtering, this performance often relies on tight sensor installation and ideal skin-electrode contact, which is often not true in real-world settings. In contrast, WISP can mitigate motion artifacts and does not require tight sensor placement.

7.3 Micro-gesture Recognition

To evaluate the micro-gesture recognition performance of our system, we conducted a user study involving 19 participants (mean age = 27.8 years).

Study Protocol. Participants were first introduced to the four micro gestures plus an additional "idle" state. Following a demonstration and practice session without the device, participants were fitted with WISP on their left wrists, positioned below the palmar carpal ligament. The standard testing position ("Arm-Z") involved

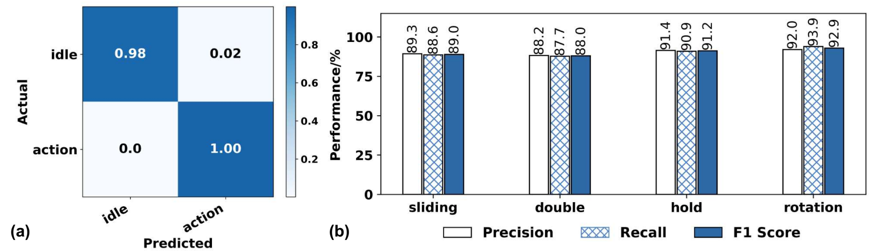


Figure 11: Micro-gesture recognition performance: (a) Hotword detection. (b) Main overall performance across four micro gestures. Results demonstrate WISP's high performance in recognizing micro gestures containing imperceptible movements.

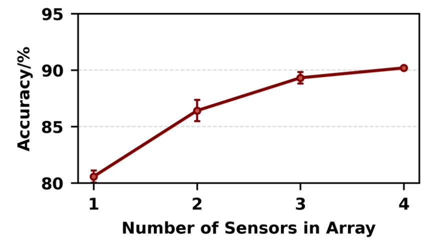


Figure 12: Performance comparison across different sensor numbers, showing the benefit of a multi-sensor array compared to a single sensor.

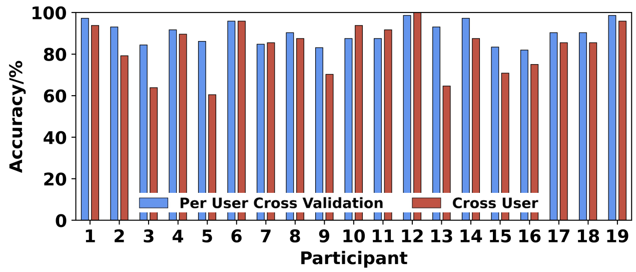


Figure 13: Per-user results for individual and cross-user model-training (with calibration), showing WISP works well for most users with only small performance degradation for cross-user study.

participants seated with their left arm perpendicular to the ground, elbow resting on a chair arm, palm oriented rightward, and wrist without clothing coverings. The study was performed in an office environment (meeting room).

Participants completed 1-2 practice rounds to familiarize themselves with the system before formal data collection. Our custom software interface randomized gesture prompts and automatically logged timing information while recording sensor data simultaneously.

Hotword Detection. We first evaluated the system's ability to distinguish between active gestures and the idle state, which serves as a "hotword detection" mechanism to activate the device from sleep mode. For this binary classification task, we grouped all four micro gestures as "action" and the idle state as "non-action." As shown in Figure 11(a), the system achieved 99.2% accuracy with minimal misclassifications in either direction, demonstrating reliable wake-word functionality.

Overall Performance. Figure 11(b) presents the precision, recall, and F1-score for the four micro-gesture classifications using 6-fold cross validation. Our system achieves 90.2% overall accuracy across all classes and users, with balanced metrics ranging between 88% and 94% for each gesture type. This shows WISP can reliably distinguish among the four base micro gestures without much bias. Among all gestures, rotation and double-press exhibit the best and worst performance, respectively, though their difference is small. Furthermore, in our study, users had no prior experience with the system and performed gestures with natural variations. We observed that performance typically improves with repeated use as users develop more consistent gesture patterns, so we expect the results to improve further with experienced users.

Cross-User Performance. Figure 13 shows the performance of training individual models with 6-fold cross-validation in blue. Nearly all users achieve an accuracy above 80%, demonstrating robust recognition of these subtle micro-gestures. To evaluate practical deployment scenarios, we extended our evaluation to include cross-user models. Specifically, we applied leave-one-user-out testing, where we tested on each user with a model trained on all other users. This approach achieved an 82.9% overall accuracy across all users with only 5 minutes of calibration data. The modest 7.3% difference between personalized and generalized models highlights WISP's ability to capture common patterns in force-based micro-gestures across different users. We also noted that while most users maintain performance comparable to individual training, Users 3, 5, and 13 show performance drops exceeding 20%. These results suggest that while our system works well out-of-the-box for most users, some individuals may benefit from additional calibration time or personalized training to achieve optimal performance. Also, we note that this cross-user evaluation captures practical challenges such as removal/reapplication of the sensor like those in a conventional cross-session study.

Multi-Channel Benefits. Figure 12 demonstrates the advantages of our multi-sensor array design. We compared the performance on all participants with a focus on different numbers of used channels. The result reveals our 4-channel array configuration outperforms single-channel by 9.3%, validating the benefit of multi-sensor array design. Additionally, the slope of the curve decreases with an increasing number of channels, indicating four channels can be a sweet spot for our design that balances efficiency and performance.

Robustness under Real-world Settings. Figure 14 demonstrates WISP's robustness across diverse real-world conditions within the same subject with 1080 randomized gesture instances and 6-fold cross-validation, showing performance differences relative to baseline conditions. The system maintains remarkably stable performance across different arm orientations (Arm-X, Arm-Y) with less than 3% variation, and shows minimal sensitivity to environmental conditions, with outdoor settings (also sitting still on a chair) actually performing slightly better than indoor lab conditions, likely due to reduced electromagnetic interference.

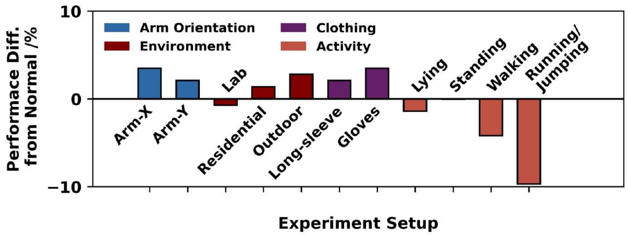


Figure 14: Sensitivity study of real-world conditions, showing WISP's robustness in various settings, even with motions. (Normal Setting: Arm-Z; Office; T-shirt; Sitting.)

Class	*S+D*	*D+S*	*S+H*	*H+S*	*S+R*	*R+S*
Precision	0.851	0.853	0.841	0.840	0.854	0.850
Recall	0.857	0.857	0.830	0.830	0.868	0.869
F1 Score	0.854	0.855	0.836	0.835	0.861	0.859
Class	*D+H*	*H+D*	*D+R*	*R+D*	*H+R*	*R+H*
Precision	0.849	0.844	0.843	0.846	0.871	0.875
Recall	0.832	0.833	0.845	0.845	0.875	0.874
F1 Score	0.840	0.838	0.844	0.845	0.873	0.874

Table 2: Performance (across all participants) for the 12 expanded micro gestures created by sequential combinations of base gestures, showing WISP's capability in reliably recognizing more complex gestures beyond the base gestures. S: sliding; D: double-press; H: hold; R: rotation.

Clothing variations reveal interesting dynamics: while basic clothing has minimal impact, long sleeves and gloves can actually enhance performance by 2-4%, likely due to increased friction and additional signal features—advantages unavailable to vision-based systems that require line-of-sight. For physical activities, WISP handles mild motions like walking with minimal degradation, though more vigorous activities like running and jumping cause approximately 10% performance drops.

Usability. Post-study, participants rated the device's wearing comfort on a 10-point scale; the mean score was 8.9/10, indicating high potential for everyday wearability.

7.4 Expanding Gesture Set

In this section, we demonstrate how the gesture set can be expanded to a larger vocabulary *without hardware modifications*.

Expanding the Dictionary with Self-Combination. To expand our gesture vocabulary beyond single gestures, we developed a more complex dataset incorporating sequential combinations of our base gestures. Specifically, we created 12 distinct gesture patterns by considering two gestures in sequence (e.g. double-press followed by press-and-hold).

With this extended gesture set, WISP achieves an average accuracy of 85.1% for all gesture combinations, across all 19 participants in our user study, as shown in Table 2. This high performance demonstrates that WISP can support a rich gesture vocabulary, enabling a wider range of sophisticated yet imperceptible interaction modes.

Expanding the Functionality with Object Interaction. WISP works not only for pinch micro-gestures, but also on various surfaces. By combining with different objects, it provides different object-specific functions. Specifically, our design involves object detection first, followed by switching the function of the four base micro gestures accordingly.

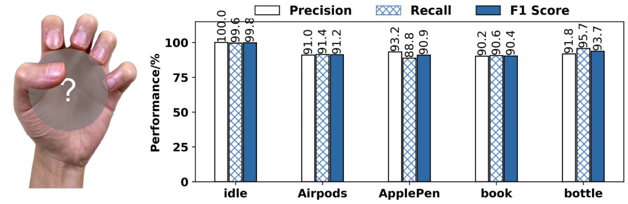
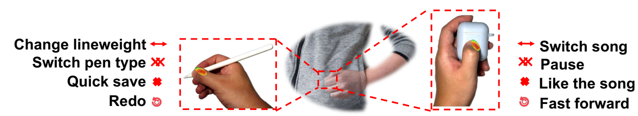


Figure 15: Object recognition performance, showing high accuracy in identifying four common objects across all participants using 6-fold cross-validation. This task is regarded as the first step to extend the functionality of WISP by making micro gestures on different object surfaces.



Performance change compared to normal (bare hands)

	Bare hands	Airpods	ApplePencil
Sliding	-	-7.2%	-1.5%
Double	-	-3.5%	+1.5%
Long	-	-0.3%	+5.6%
Rotation	-	+2.8%	+2.8%

Figure 16: Performance on different object surfaces in a clothes pocket, enabling extended functionality solely from the four base micro-gestures, even without line-of-sight conditions.

For the first stage, object recognition, we conducted a new user study with 13 participants using a similar setup to our main study. The study includes four common objects: Airpods case, Apple Pencil, book, and bottle. Figure 15 shows a promising overall accuracy of 93.8% using 6-fold cross-validation. The underlying principle is also similar to micro gesture recognition: when interacting with different objects, our tendons and skin at the wrist deform differently, making these interactions detectable.

Secondly, we compared the performance differences when making micro gestures on different objects. As Figure 16 shows, we selected an AirPods case and an Apple Pencil since they are small enough to put in the pocket, preserving the imperceptible feature. The gestures were mapped to different functions for each object (e.g., music control for AirPods, pen settings for Apple Pencil).

The user was asked to make micro gestures with these two objects in the pocket in our preliminary study, involving a non-line-of-sight scenario with restricted space. The results indicate their performance is comparable to bare-hand interaction, suggesting that the underlying micro-gesture mechanics remain similar across different surfaces. This method (objects × base gestures) can significantly extend the number of gestures. Furthermore, this expansion can be easily achieved with WISP, without any hardware modification.

7.5 System Benchmarks

Hardware Solution Comparison. We now evaluate the contribution of key hardware components to gesture recognition

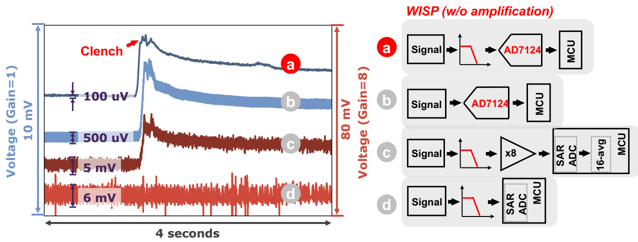


Figure 17: Contribution of hardware stages to removing signal noise. (a) WISP setting. (b) Bypass the analog LPF in the circuit. (c) Use an MCU-integrated SAR ADC with amplifier gain of 8-fold and 16-point averaging. (d) Use an MCU-integrated SAR ADC integrated without any signal conditioning.

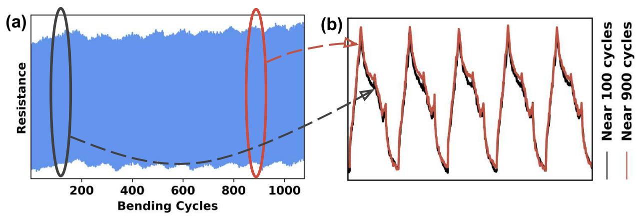


Figure 18: Sensor stability under repeated endings: (a) The resistance change under over 1000 bending cycles. (b) Zoom-in result near 100 bendings and near 900 bendings, showing a highly consistent response under intensive bendings.

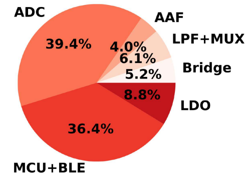
Classifier		RF	SVM (RBF)	KNN	2-layer MLP
Features	WISP	**90.2**	89.4	84.9	89.0
	Stat.	75.0	76.5	72.7	79.3

Table 3: Classification accuracy (%) comparison across different classifiers and feature extraction methods.

performance using raw data without software processing to ensure fair comparison. Figure 17(a-b) shows that the analog LPF proves critical — removing this module resulted in 5x more noise ($100\mu Vpp \rightarrow 500\mu Vpp$). Figure 17(c-d) shows that substituting our 24-bit ADC with a 12-bit MCU-integrated SAR ADC (Teensy 3.2) can lead to much more noise ($6mVpp$), which drowns out the useful signal. Even with 8× amplification and 16-point averaging, the SAR ADC cannot match WISP's signal quality. These results validate our hardware design choices for sub-10mm gesture detection.

Sensor Robustness to Repeated Bending. We now evaluate the robustness of WISP to repeated use. An automatic force gauge with motors [8] was employed for repeatable sensor bending. This bending is significantly stronger than that typically experienced on the wrist, representing a worst-case scenario. Figure 18(a) shows the sensor remains reliable after 1000 hard bending cycles. A zoomed-in view of the early stages near 100 bendings and later stages near 900 bendings, as shown in Figure 18(b), presents a highly consistent response even after extensive bending.

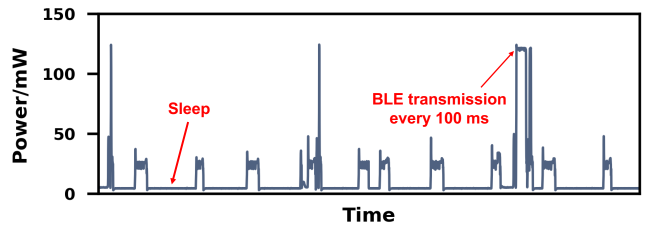
Feature Extraction Analysis. Table 3 presents a comparison of different classifiers and feature extraction methods. Our results show that the combination of Random Forest classifier (RF) with the Catch22 + KPCA features provides the best performance, achieving 90.2% accuracy. This significantly outperforms simpler statistical



(a)

Component	Avg power	Mode
Bridge	0.53 mW	always on
LPF+MUX	0.62 mW	always on
AAF	0.41 mW	always on
ADC	3.99 mW	duty-cycle
MCU+BLE	3.69 mW	duty-cycle
LDO	0.89 mW	always on
Total	10.13 mW	/

(b)



(c)

Figure 19: Power consumption profile of WISP. (a) Percentage breakdown of power consumption (%). (b) Average power consumption and operating mode of each component. (c) Real-time, continuous power monitoring of the entire system.

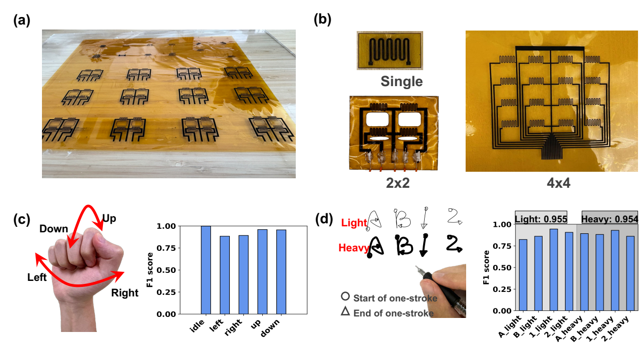


Figure 20: Extension of WISP in fabrication and application. (a) One-stop batch printing of of heterogeneous patterns. (b) Sensor arrays of different configurations. (c) Wrist rotational movements. (d) Pressure-sensing writing, where wrist-based sensing could distinguish handwriting with different characters and force levels (light or heavy).

features (max, mean, STD, quantiles, RSSQ (Root-sum-of-squares level)), demonstrating the value of using time-series features for capturing the nuanced characteristics of our sensor signals.

Power Consumption Analysis. A detailed power profiling using the Nordic NRF PPK2 tool [43] is shown in Figure 19. Our system achieves excellent signal quality ($100\mu Vpp$ noise floor) while maintaining low power consumption (10.13 mW total). Using a 1000mAh-level LiPo battery (e.g. [14, 28]) of a similar size to our circuit ($4cm \times 4cm$), our system can run for around two weeks without charging.

8 Discussion and Conclusion

The scope of WISP can be expanded further in both fabrication and application aspects.

Fabrication Flexibility. While WISP utilizes a 2x2 array with a zigzag pattern, we have extended it to larger arrays (Figure 20(a)), and a wide range of array configurations (Figure 20(b)). The ability to print a diversity of patterns can potentially be used to improve WISP's performance and to scale up the system.

Other Use Cases. WISP can readily be extended to other applications. Figure 20(c) presents the user study results for wrist rotational movements (left, right, up, and down), which leverage the same force-based sensing principle but involve larger movements than micro-gestures. The system achieves 95.2% average accuracy across 13 participants using 6-fold cross-validation. Figure 20(d) demonstrates a novel application where our sensor transforms conventional pen-and-paper writing into a pressure-sensitive input modality, similar to an iPad. In a preliminary study with four characters (A, B, 1, 2) written at explicitly different pressure levels (light and heavy), the system achieved 88.8% accuracy in distinguishing all eight combinations.

The Scope of Hotword Detection. A fully deployed real-world gesture recognition system would require three stages: (1) distinguishing everyday activities from micro-gesture states, (2) detecting gesture initiation from "idle" state inside the micro-gesture set, and (3) classifying specific micro gestures. This work focuses on stages (2) and (3) as they present novel technical challenges for imperceptible gesture recognition. While stage (1) is important to prevent false positives from activities such as typing, it resembles coarser-grained gesture recognition tasks addressed in prior work and can leverage well-established solutions with high accuracy.

Lifespan, Diversity, and Social Perceptibility. This work has evaluated WISP's performance across multiple dimensions: multi-user studies, diverse real-world settings, object interaction, and technical benchmarks including SNR, sensor longevity, and power consumption. To further validate real-world usability, future work should include longitudinal deployment studies, evaluation with diverse populations, environmental impact assessment of disposable sensors, and perceptual validation of visual imperceptibility from observer perspectives.

Conclusion. In conclusion, we proposed a novel force-based micro-gesture set, characterized by extremely minimal movements. To distinguish these gestures solely from the wrist, we explored a novel sensing paradigm that integrates batch fabrication of a 2D Laser-Induced Graphene (LIG) array sensor with a high-performance, low-power hardware-software co-design. These measures enabled our system to simultaneously achieve low cost, high comfort, and high performance. We also showed that our system is robust under various real-world interference and can work with expanded gesture sets with human-object interaction.

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